

## ILE DE BREHAT, France

WMO: 071210

Lat: 48.856N

Lon: 3.004W

Elev: 118

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 31.6       | 34.2 | 23.2                          | 17.6 | 34.0 | 26.5 | 20.5 | 35.9 | 44.9                  | 47.2 | 41.8 | 46.5 | 15.0                  | 80   | 0.681 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.7                    | 72.0            | 64.8 | 69.9 | 64.0 | 68.2 | 63.1 | 66.3                | 70.1 | 65.0 | 68.4 | 63.9 | 67.1 | 11.2                 | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.8                            | 92.1 | 67.9 | 63.7 | 88.6 | 66.8 | 62.6 | 85.3 | 65.8 | 31.1          | 70.6 | 30.0 | 68.2 | 29.2 | 67.4 | 72.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 34.5              | 30.1 | 27.0 | DB                         | 26.6 | 79.1               | 3.4 | 3.1                                                | 24.2 | 81.3       | 22.2 | 83.1       | 20.3 | 84.8       | 17.8 | 87.1 |
|                   |      |      | WB                         | 24.4 | 69.0               | 3.7 | 1.6                                                | 21.8 | 70.2       | 19.6 | 71.1       | 17.6 | 72.0       | 14.9 | 73.2 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 53.4   | 45.0  | 44.8  | 47.8  | 49.3  | 54.4  | 58.3  | 62.5  | 63.4  | 60.5  | 56.3  | 50.5  | 47.2 |
|                                                                                  | DBStd       | 7.64   | 5.45  | 5.53  | 3.79  | 3.70  | 3.67  | 3.15  | 2.69  | 2.99  | 3.26  | 3.85  | 4.46  | 4.84 |
|                                                                                  | HDD50       | 628    | 164   | 156   | 88    | 55    | 5     | 0     | 0     | 0     | 0     | 2     | 48    | 110  |
|                                                                                  | HDD65       | 4276   | 619   | 567   | 534   | 472   | 330   | 203   | 86    | 68    | 141   | 269   | 436   | 551  |
|                                                                                  | CDD50       | 1861   | 10    | 9     | 19    | 33    | 140   | 249   | 389   | 414   | 314   | 199   | 61    | 24   |
|                                                                                  | CDD65       | 34     | 0     | 0     | 0     | 0     | 0     | 2     | 10    | 17    | 5     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 37     | 0     | 0     | 0     | 0     | 1     | 5     | 9     | 21    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 1      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 13.9   | 16.3  | 16.1  | 14.5  | 14.1  | 12.3  | 12.2  | 11.7  | 11.3  | 12.8  | 14.5  | 15.1  | 16.0 |
| Precipitation                                                                    | PrecAvg     | 32.6   | 3.4   | 3.1   | 2.1   | 2.5   | 2.2   | 1.9   | 1.9   | 2.0   | 1.7   | 3.3   | 4.0   | 4.0  |
|                                                                                  | PrecMax     | 40.8   | 6.5   | 6.1   | 6.3   | 6.9   | 4.7   | 4.4   | 5.0   | 3.9   | 5.1   | 6.9   | 7.9   | 8.8  |
|                                                                                  | PrecMin     | 24.6   | 0.7   | 0.7   | 0.2   | 0.7   | 0.6   | 0.4   | 0.7   | 0.7   | 0.1   | 0.5   | 1.5   | 0.9  |
|                                                                                  | PrecStd     | 3.8    | 1.7   | 1.5   | 1.3   | 1.4   | 0.9   | 1.1   | 1.0   | 0.9   | 1.1   | 1.5   | 1.7   | 1.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 56.6  | 58.1  | 60.3  | 64.4  | 68.1  | 71.9  | 74.5  | 76.8  | 72.4  | 68.9  | 61.0  | 58.5 |
|                                                                                  |             | MCWB   | 53.9  | 55.0  | 55.4  | 57.2  | 61.3  | 63.8  | 65.2  | 67.0  | 64.7  | 62.3  | 57.9  | 55.5 |
|                                                                                  | 2%          | DB     | 54.9  | 55.3  | 56.9  | 59.6  | 64.9  | 67.9  | 71.1  | 72.4  | 69.1  | 65.3  | 59.1  | 57.0 |
|                                                                                  |             | MCWB   | 52.7  | 52.5  | 53.0  | 54.2  | 59.0  | 62.8  | 64.8  | 65.1  | 63.7  | 60.6  | 56.7  | 54.5 |
|                                                                                  | 5%          | DB     | 53.9  | 53.6  | 54.8  | 56.6  | 62.5  | 64.8  | 69.0  | 70.2  | 67.1  | 63.0  | 57.6  | 55.7 |
|                                                                                  |             | MCWB   | 51.9  | 51.1  | 51.7  | 52.9  | 57.9  | 60.3  | 63.8  | 64.3  | 62.4  | 59.2  | 55.5  | 53.5 |
|                                                                                  | 10%         | DB     | 52.1  | 52.1  | 53.1  | 54.7  | 60.1  | 63.1  | 67.3  | 68.4  | 65.4  | 61.3  | 56.3  | 54.2 |
|                                                                                  |             | MCWB   | 50.3  | 50.0  | 50.5  | 51.7  | 56.2  | 59.2  | 63.0  | 63.1  | 61.3  | 58.3  | 54.2  | 52.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 54.6  | 55.8  | 56.0  | 57.8  | 62.6  | 65.4  | 68.0  | 68.3  | 66.3  | 63.2  | 58.9  | 57.0 |
|                                                                                  |             | MCDB   | 55.7  | 57.3  | 59.3  | 62.8  | 66.6  | 69.9  | 71.7  | 73.5  | 69.7  | 66.8  | 60.0  | 58.1 |
|                                                                                  | 2%          | WB     | 53.5  | 53.3  | 53.9  | 55.1  | 59.8  | 62.9  | 65.9  | 66.7  | 64.7  | 61.3  | 57.2  | 55.3 |
|                                                                                  |             | MCDB   | 54.8  | 54.7  | 56.2  | 58.2  | 63.5  | 66.8  | 69.2  | 71.0  | 67.8  | 63.8  | 58.3  | 56.3 |
|                                                                                  | 5%          | WB     | 52.2  | 51.6  | 52.2  | 53.5  | 58.2  | 61.1  | 64.5  | 65.3  | 63.6  | 60.2  | 56.1  | 54.0 |
|                                                                                  |             | MCDB   | 53.5  | 53.1  | 54.1  | 56.0  | 61.6  | 64.0  | 67.8  | 68.9  | 66.4  | 62.2  | 57.4  | 55.5 |
|                                                                                  | 10%         | WB     | 50.4  | 50.2  | 50.9  | 52.1  | 56.7  | 59.7  | 63.4  | 64.2  | 62.2  | 59.0  | 54.8  | 52.4 |
|                                                                                  |             | MCDB   | 52.0  | 51.9  | 52.8  | 54.2  | 59.8  | 62.3  | 66.5  | 67.2  | 64.6  | 60.8  | 56.3  | 54.1 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.0   | 6.5   | 7.5   | 7.5   | 8.2   | 8.0   | 8.2   | 8.7   | 7.4   | 6.9   | 6.2   | 6.0  |
|                                                                                  |             | MCDBR  | 7.4   | 8.4   | 10.0  | 10.7  | 12.5  | 11.2  | 11.0  | 11.6  | 9.5   | 9.1   | 7.3   | 7.5  |
|                                                                                  |             | MCWBR  | 6.6   | 7.3   | 7.6   | 7.0   | 7.5   | 6.4   | 6.2   | 6.0   | 5.6   | 6.0   | 6.2   | 6.7  |
|                                                                                  | 5% WB       | MCDBR  | 7.1   | 7.9   | 9.2   | 10.0  | 11.4  | 10.9  | 9.9   | 10.2  | 8.8   | 8.2   | 6.6   | 6.9  |
|                                                                                  |             | MCWBR  | 7.0   | 7.8   | 7.4   | 7.0   | 7.5   | 6.6   | 6.2   | 6.1   | 5.9   | 6.2   | 6.3   | 6.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333  | 0.349 | 0.378 | 0.390 | 0.395 | 0.389 | 0.387 | 0.377 | 0.366 | 0.359 | 0.336 | 0.326 |      |
|                                                                                  | taud        | 2.414  | 2.393 | 2.322 | 2.311 | 2.347 | 2.392 | 2.400 | 2.447 | 2.459 | 2.465 | 2.445 | 2.382 |      |
|                                                                                  | Ebn at Noon | 228    | 252   | 263   | 272   | 275   | 277   | 275   | 273   | 265   | 246   | 226   | 212   |      |
|                                                                                  | Edh at Noon | 21     | 27    | 34    | 37    | 38    | 36    | 36    | 33    | 29    | 25    | 20    | 19    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 295    | 540   | 898   | 1407  | 1679  | 1840  | 1763  | 1488  | 1163  | 668   | 363   | 246   |      |
|                                                                                  | RadStd      | 24     | 63    | 93    | 129   | 132   | 195   | 103   | 118   | 84    | 52    | 35    | 30    |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page